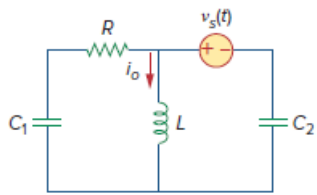


Problem

Using Fig. 10.108, design a problem to help other students better understand Norton's theorem.

Figure 10.108:



Step-by-step solution

Step 1 of 9

Consider the following circuit diagram:

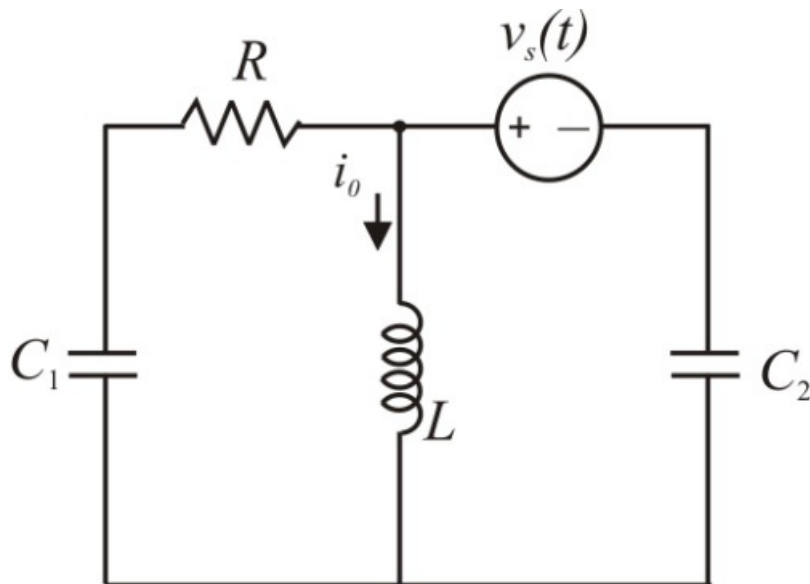


Figure 1

Step 2 of 9

Assume the network parameter values and consider the following circuit:

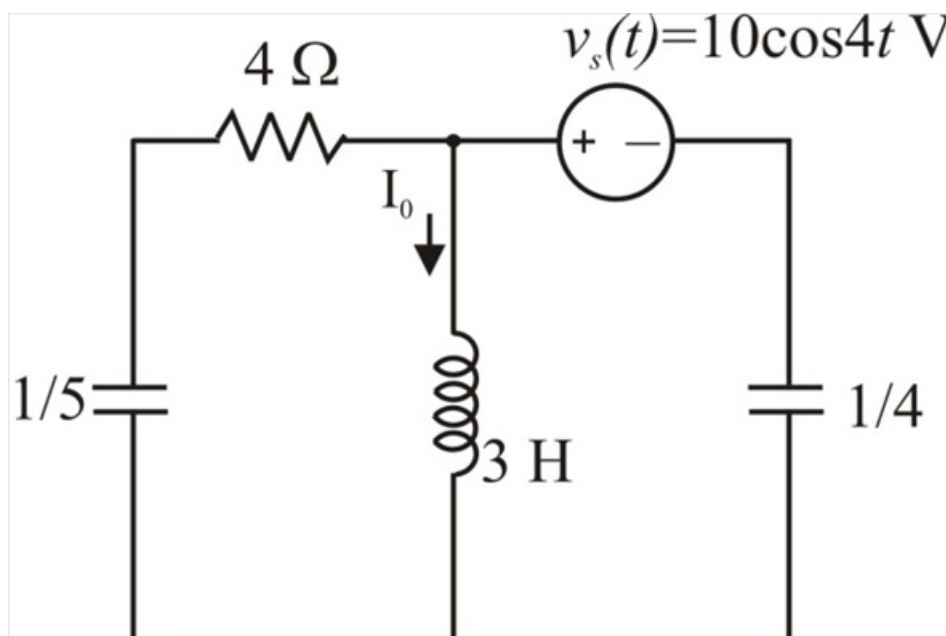


Figure 2

Step 3 of 9

Convert the circuit in Figure 1 to the frequency domain:

$$v_s(t) = 10 \cos 2t \text{ V}$$

$$v_s = 10 \angle 0^\circ \text{ V}$$

And

$$\omega = 4 \text{ rad/s}$$

Convert the inductance value.

$$L = 3 \text{ H}$$

$$\begin{aligned} j\omega L &= j(4 \text{ rad/s})(3 \text{ H}) \\ &= j12 \Omega \end{aligned}$$

Convert the capacitance value.

$$C_1 = \frac{1}{5} \text{ F}$$

$$\begin{aligned} \frac{1}{j\omega C_1} &= \frac{1}{j(4 \text{ rad/s})\left(\frac{1}{5} \text{ F}\right)} \\ &= \frac{1}{j(0.8)} \Omega \\ &= -j1.25 \Omega \end{aligned}$$

Convert the capacitance value.

$$C_2 = \frac{1}{4} \text{ F}$$

$$\begin{aligned} \frac{1}{j\omega C_2} &= \frac{1}{j(4 \text{ rad/s})\left(\frac{1}{4} \text{ F}\right)} \\ &= \frac{1}{j(1)} \Omega \\ &= -j \Omega \end{aligned}$$

Step 4 of 9

Draw the equivalent frequency domain circuit:

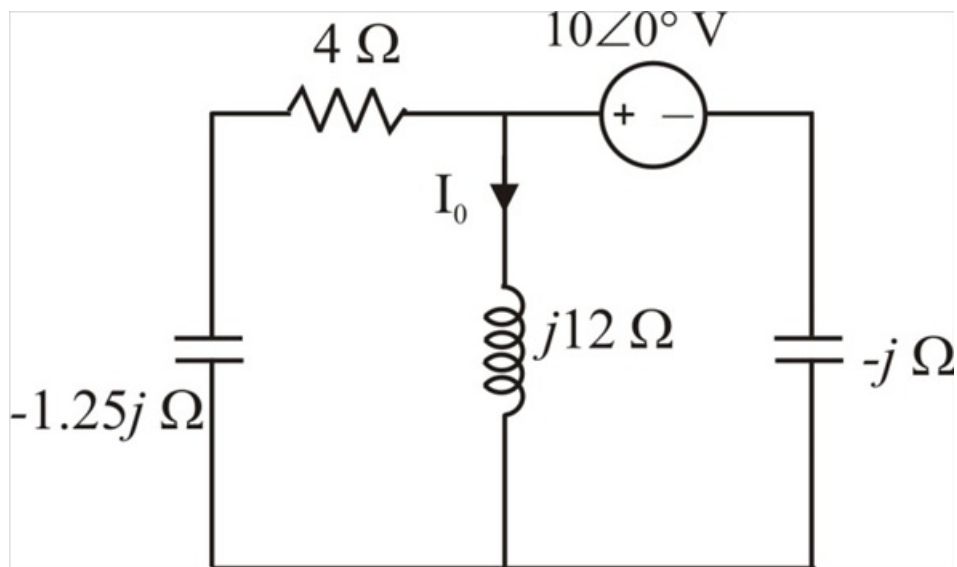


Figure 3

Step 5 of 9

Consider the following circuit to find the Norton's equivalent impedance $\mathbf{Z_N}$:

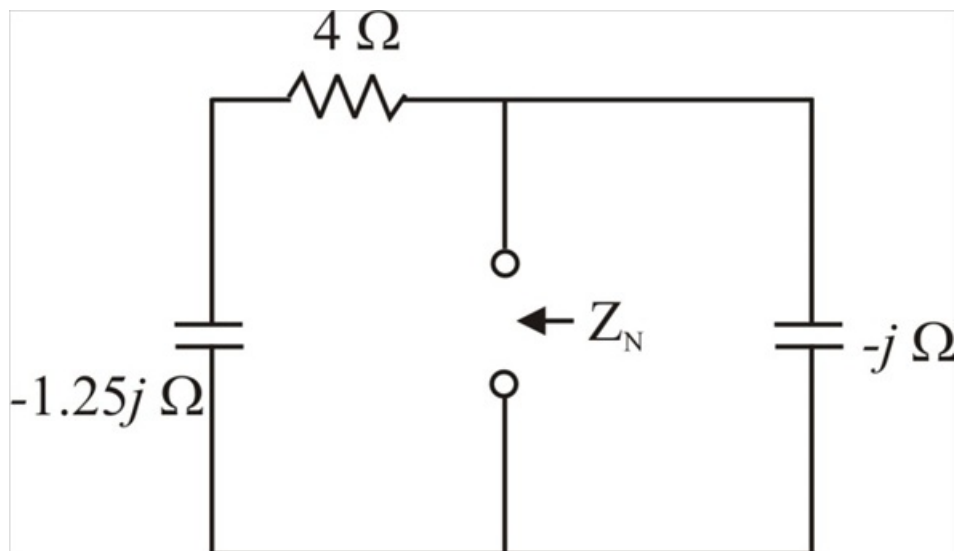


Figure 4

Calculate the value of $\mathbf{Z_N}$:

$$\begin{aligned}\mathbf{Z_N} &= (-j) \parallel (4 - j1.25) \\ &= \frac{(-j)(4 - j1.25)}{-j + 4 - j1.25} \\ &= 0.1899 - j0.893\end{aligned}$$

Step 6 of 9

Consider the following circuit to find the Norton's equivalent current, $\mathbf{I_N}$:

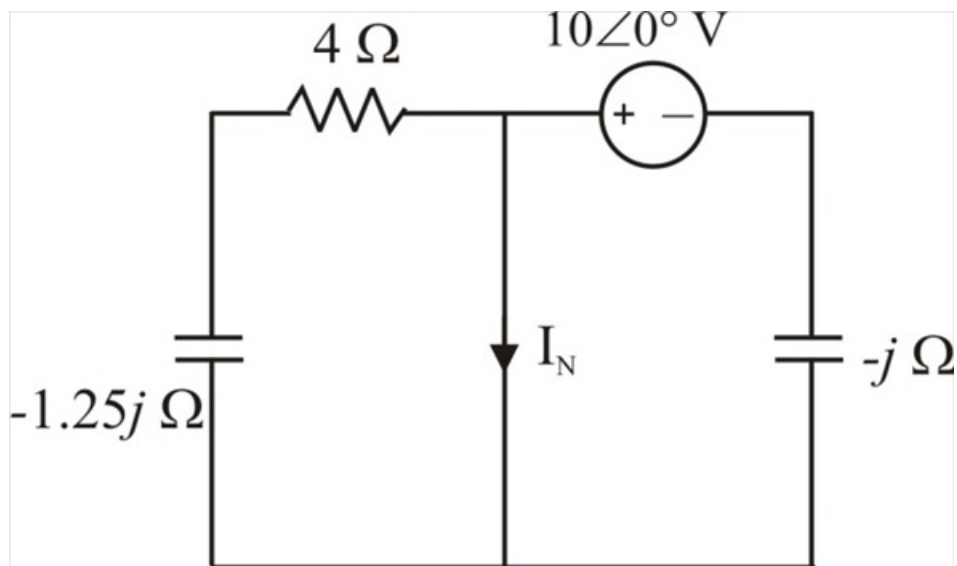


Figure 5

Step 7 of 9

Calculate the value of $\mathbf{I_N}$:

$$\begin{aligned}\mathbf{I_N} &= \frac{10\angle 0^\circ \text{ V}}{-j} \\ &= j10 \text{ A}\end{aligned}$$

Step 8 of 9

Draw the Norton's equivalent circuit:

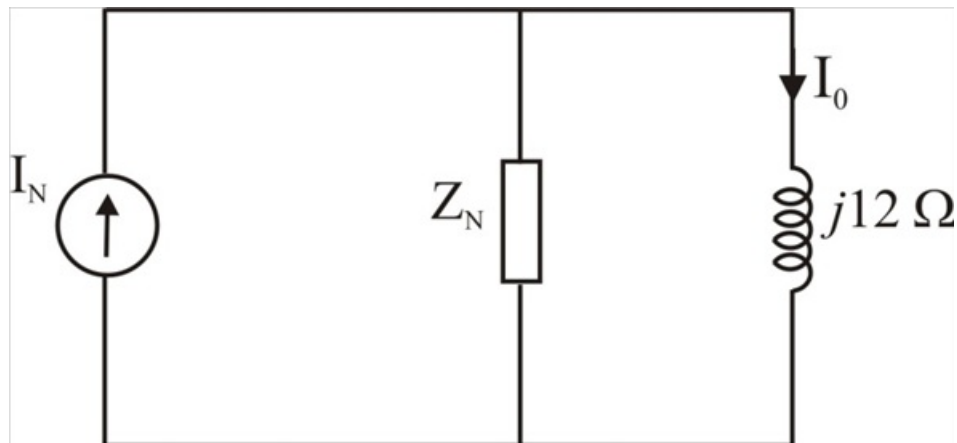


Figure 6

Step 9 of 9

Use the current division method to calculate the value of $\mathbf{I}_o$:

$$\begin{aligned}\mathbf{I}_o &= \frac{\mathbf{Z}_N}{\mathbf{Z}_N + j12} (\mathbf{I}_N) \\ &= \frac{0.1899 - j0.893}{0.1899 - j0.893 + j12} (j10) \\ &= 0.1846 - j0.8008 \\ &= 0.821 \angle -77.015^\circ \text{ A}\end{aligned}$$

Therefore the value of $\mathbf{I}_o$ is $\boxed{821 \cos(4t - 77.015^\circ) \text{ mA}}$.